A weak*-to-weak continuous operator on ℓ^1 with uncomplemented kernel

Agent packet for `fa_banach_001`

2026-06-28

Source and target question

The source paper is Flemming–Hofmann, *New aspects of ill-posedness classification in Banach spaces*, arXiv:2511.06690 [1]. After proving that Mazur-type operators and their nonzero compositions fail weak*-to-weak continuity, the authors write that they were unable to find weak*-to-weak continuous operators on ℓ^1 with uncomplemented null-space. They then ask whether, in ℓ^1, weak*-to-weak continuity implies complementedness of the null-space.

structure of the minimizers provided by the proposition above one easily sees that the sequence of minimizers always converges weakly* to 0 (and it is not norm-Cauchy).

The authors conjecture that in ℓ^1 there is an intimate connection between both lacking weak*-to-weak continuity and uncomplemented null-spaces, driven by the fact that they were unable to find weak*-to-weak continuous operators with uncomplemented null-space. Both classes of operators with uncomplemented null-space, Mazur-type operators as well as quotient maps, are not weak*-to-weak continuous by Theorem 3.2. The following theorem shows that even compositions with Mazur-type operators (cf. Examples 2.7 and 2.8) always lack weak*-to-weak continuity.

Theorem 3.4. *Let $A_{\mathrm{Maz}} : \ell^1 \to Y$ be of Mazur-type with Y some separable Banach space (cf. Example 2.5) and let $C : Y \to Z$ with $C \neq 0$ be some bounded linear operator mapping into a Banach space Z. Then $A := C \circ A_{\mathrm{Maz}}$ is not weak*-to-weak continuous.*

Proof. For $n \in \mathbb{N}$ consider $e^{(n)}$ with 1 at position n and zeros else. The sequence $(e^{(n)})_{n\in\mathbb{N}}$ converges weakly* to zero in ℓ^1. We show that there is some $\eta \in Z^*$ such that the dual product $\langle \eta, A\,e^{(n)}\rangle_{Z^*,Z}$ does not tend to zero as $n \to \infty$, which would mean that $(A\,e^{(n)})_{n\in\mathbb{N}}$ does not converge weakly to zero in Z and would prove the theorem. Now, for arbitrary $\eta \in Z^*$, we have

$$\langle \eta, A\,e^{(n)}\rangle_{Z^*,Z} = \sum_{k=1}^{\infty} e_k^{(n)} \langle \eta, C\,\zeta^{(k)}\rangle_{Z^*,Z} = \langle \eta, C\,\zeta^{(n)}\rangle_{Z^*,Z} = \langle C^*\eta, \zeta^{(n)}\rangle_{Y^*,Y}. \tag{3.23}$$

Take some η with $C^*\eta \neq 0$ and some $y \in Y$ with $\langle C^*\eta, y\rangle_{Y^*,Y} \neq 0$ and $\|y\|_Y = 1$. Then we find a sequence $(k_n)_{n\in\mathbb{N}}$ such that there is norm convergence $\zeta^{(k_n)} \to y$ in Y as $n \to \infty$. Thus

$$\langle C^*\eta, \zeta^{(k_n)}\rangle_{Y^*,Y} \to \langle C^*\eta, y\rangle_{Y^*,Y} \neq 0, \tag{3.24}$$

and the proof is complete. □

Weak*-to-weak continuity would imply weak* closedness of the null-space, which in non-reflexive Banach spaces is a stronger property than norm or weak closedness (note that a subspace is norm closed if and only if it is weakly closed). Does weak* closedness of the null-space imply its complementedness in ℓ^1? In reflexive Banach spaces weak* and weak convergence

coincide and all bounded linear operators are weak*-to-weak continuous. Neththeless, complemented as well as uncomplemented null-spaces may occur and there is no relationship between complementedness of null-spaces and weak*-to-weak continuity. In ℓ^∞ it is known that uncomplemented null-spaces may occur even for weak*-to-weak continuous operators, see [15, Theorem 3.1]. In ℓ^1 the question whether weak*-to-weak continuity implies complementedness of the null-space remains open. At least the converse is not true. The following example shows that there are ill-posed operators with complemented null-space which are not weak*-to-weak continuous.

Result

We give a counterexample. There is a bounded operator

$$A : \ell^1 \longrightarrow c_0$$

which is weak*-to-weak continuous, but $\ker A$ is not complemented in ℓ^1. In particular, weak* closedness of the null-space also does not imply complementedness in ℓ^1.

Proof intuition

Weak*-to-weak continuity of an operator $A : \ell^1 = c_0^* \to Y$ is easy to force if every scalar functional on the range pulls back to an element of c_0. Thus we build A from scalar coordinates indexed by a dense sequence in a subspace $E \subset c_0$; then $A^*(Y^*) \subset E \subset c_0$ and $\ker A = E^\perp$.

The remaining task is to choose E so that $E^\perp$ is not complemented in ℓ^1. If $E^\perp$ were complemented, then the quotient $\ell^1/E^\perp \cong E^*$ would be a complemented subspace of ℓ^1, hence isomorphic to ℓ^1. We choose

$$E \cong \Big(\bigoplus_{n=1}^{\infty} \ell_2^n\Big)_{c_0}.$$

Its dual is the ℓ^1-sum of the Hilbert blocks ℓ_2^n, which is not isomorphic to ℓ^1 because its dual contains uniformly 1-complemented Hilbert blocks of unbounded dimension.

Auxiliary facts

Lemma 1. *The space*

$$E_0 = \Big(\bigoplus_{n=1}^{\infty} \ell_2^n\Big)_{c_0}$$

embeds isomorphically into c_0, while

$$E_0^* = \Big(\bigoplus_{n=1}^{\infty} \ell_2^n\Big)_{\ell^1}$$

is not isomorphic to ℓ^1.

Proof. For each n, choose a finite 1/2-net F_n in the unit sphere of ℓ_2^n. The map

$$J_n h = (\langle h, u\rangle)_{u\in F_n} \in \ell^\infty(F_n)$$

satisfies

$$\frac{1}{2}\|h\|_2 \le \|J_n h\|_\infty \le \|h\|_2.$$

The block diagonal map $\oplus_n J_n$ embeds E_0 into the finite-block c_0-sum $(\oplus_n \ell^\infty(F_n))_{c_0}$, which is naturally isometric to c_0.

The dual formula follows from the standard duality of c_0-sums of finite-dimensional spaces. To see that this dual is not isomorphic to ℓ^1, suppose otherwise. Then its dual

$$\Big(\bigoplus_{n=1}^{\infty} \ell_2^n\Big)_{\ell^\infty}$$

would be isomorphic to ℓ^∞. Since ℓ^∞ is injective, this would make $(\oplus_n \ell_2^n)_{\ell^\infty}$ λ-injective for some finite λ. Each coordinate Hilbert block ℓ_2^n is 1-complemented in that space, so the injectivity constants of the spaces ℓ_2^n would be bounded independently of n. This contradicts the classical fact that the projection/injectivity constants of finite-dimensional Hilbert spaces tend to infinity with n; in fact they are of order $\sqrt{n}$ [2, 3]. □

Lemma 2. *Let $E \subset c_0$ be a closed subspace such that E^* is not isomorphic to ℓ^1. Then its annihilator*

$$E^\perp = \{x \in \ell^1 : \langle x, e\rangle = 0 \text{ for all } e \in E\}$$

is not complemented in ℓ^1.

Proof. The quotient map $\ell^1 = c_0^* \to E^*$ has kernel $E^\perp$ and is onto by Hahn–Banach, so $\ell^1/E^\perp \cong E^*$. If $E^\perp$ were complemented in ℓ^1, then E^* would be isomorphic to a complemented subspace of ℓ^1. By the classical Pełczynski theorem, every infinite-dimensional complemented subspace of ℓ^1 is isomorphic to ℓ^1 [3]. This contradicts the hypothesis on E^*. □

Counterexample

Theorem 1. *There is a bounded weak*-to-weak continuous operator $A : \ell^1 \to c_0$ whose kernel is uncomplemented in ℓ^1.*

Proof. Let $E \subset c_0$ be an isomorphic copy of $E_0 = (\oplus_n \ell_2^n)_{c_0}$ as in Lemma 1. Choose a sequence $(v_k)_{k=1}^\infty$ in the unit ball of E whose linear span is dense in E. Define

$$Ax = \left(2^{-k}\langle x, v_k\rangle\right)_{k=1}^\infty, \qquad x \in \ell^1 = c_0^*.$$

This is a bounded operator from ℓ^1 into c_0, since $|2^{-k}\langle x, v_k\rangle| \le 2^{-k}\|x\|_1$.

We verify weak*-to-weak continuity. Let $a = (a_k) \in \ell^1 = c_0^*$. Then for $x \in \ell^1$,

$$\langle a, Ax\rangle = \sum_{k=1}^\infty a_k 2^{-k}\langle x, v_k\rangle = \left\langle x, \sum_{k=1}^\infty a_k 2^{-k} v_k \right\rangle.$$

The series $\sum_k a_k 2^{-k} v_k$ converges in norm in $E \subset c_0$. Hence $A^*a \in c_0$ for every $a \in \ell^1$. Equivalently, every scalar functional on c_0 pulls back to a weak*-continuous functional on ℓ^1. Therefore A sends weak*-convergent nets, and in particular weak*-convergent sequences, in ℓ^1 to weakly convergent nets in c_0.

Finally,

$$\ker A = \{x \in \ell^1 : \langle x, v_k\rangle = 0 \text{ for all } k\} = E^\perp,$$

because the span of (v_k) is dense in E. By Lemmas 1 and 2, $E^\perp$ is not complemented in ℓ^1. □

Verification and limitations

The construction uses no numerical search. The formula for A directly gives boundedness, weak*-to-weak continuity through the adjoint inclusion $A^*(\ell^1) \subset c_0$, and $\ker A = E^\perp$. The only external Banach space facts used are standard: finite-dimensional Hilbert spaces have unbounded injectivity/projection constants, ℓ^∞ is injective, and every infinite-dimensional complemented subspace of ℓ^1 is isomorphic to ℓ^1.

The counterexample answers the source question negatively in the sequential sense used in the paper, and actually gives the stronger net-continuity version. It also refutes the intermediate question whether weak* closedness of the null-space implies complementedness in ℓ^1.

Novelty check

The run's lightweight indexes had no existing packet for arXiv:2511.06690. Searches were run for exact and close phrases including:

- `weak*-to-weak continuity complemented ell^1 null-space`;
- `weak* closed subspace ell^1 complemented`;
- `weak*-closed subspace l_1 uncomplemented`.

No direct later answer to the Flemming–Hofmann question was found. The result should nevertheless be reviewed as a candidate counterexample because it relies on classical local-theory facts rather than on a cited paper explicitly answering this 2025 question.

Human review recommendation

Recommended review focus: check the standard local-theory step in Lemma 1, namely that $(\oplus_n \ell_2^n)_{\ell^1}$ is not isomorphic to ℓ^1 via the unbounded projection constants of ℓ_2^n. The operator construction and kernel computation are otherwise direct.

References

[1] J. Flemming and B. Hofmann, *New aspects of ill-posedness classification in Banach spaces*, arXiv:2511.06690, 2025.

[2] B. Grunbaum, *Projection constants*, Trans. Amer. Math. Soc. 95 (1960), 451–465.

[3] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I. Sequence Spaces*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 92, Springer-Verlag, 1977.